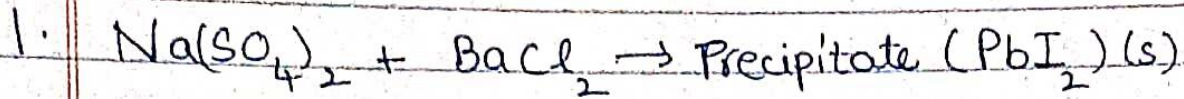


Name: Ananya Jere - Batch - BIONISD
CHEMISTRY

Calcium
Magnesium
Aluminium
Chlorine
Zinc



The precipitate formed is in Yellow colour.

Iron
Tin
Lead

2. ~~H_2SO_4~~ Ammonia gas is evolved. (NH_3).

Hydrogen
Copper

3. Assertion is True but Reason is False.

Silver
Gold

5. Assertion is True but Reason is False.

6. In the cases of Aluminium and Zinc strips, the student would observe a displacement reaction as, these both are more reactive than FeSO_4 and hence will displace the Fe.

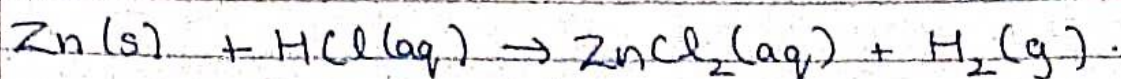
7. On heating green vitriol crystals, a reddish brown solid is left out and the smell of gas which gives off a burning Sulphur is ~~observed~~ obtained. This is Thermal Decomposition.

8.

(a) Zinc Granules.

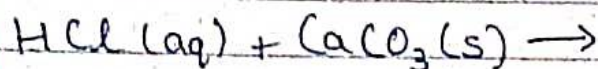
Here, Zinc displaces Hydrogen as it is more reactive and will produce Hydrogen.

gas and Zinc chloride



(b) Crushed egg shells.

In this process / reaction, dilute HCl reacts with crushed eggshells that are made of Calcium Carbonate. Hence, making the shells softer.

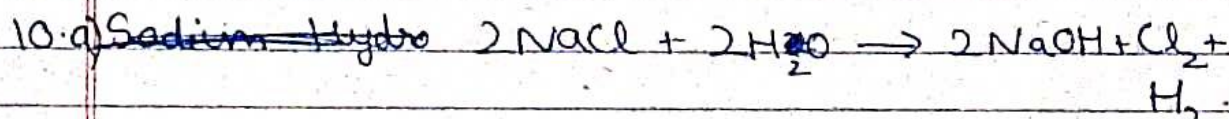


9. The pH of a salt used to make crispy pakoras is 14. The salt type is Baking Soda i.e., Sodium bicarbonate (NaHCO_3).

The uses of Baking Soda are:-

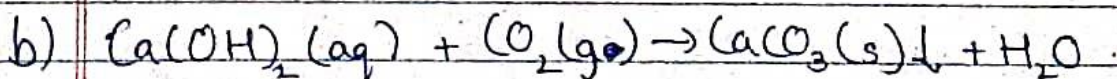
* It is used as an Antacid as it has the properties to reduce Acidity in the Stomach.

* It also is used for cooking fried foods like Pakoras.



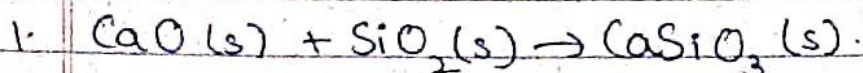
The process is called Chloro-Alkali.

In this process Hydrogen gas is evolved at Cathode & Chlorine gas is evolved at Anode.



When excess Carbon Dioxide is added, it reacts with CaCO_3 & H_2O to form Calcium bicarbonate & since it is soluble in water, it dissolves making the lime water colourless.

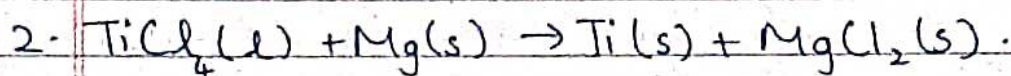
11.



	R	P
Ca	1	1
O	2	3
Si	1	1

∴ The balanced eqⁿ is $\text{CaO (s)} + \text{SiO}_2\text{(s)} \rightarrow \text{CaSiO}_3\text{(s)}$.

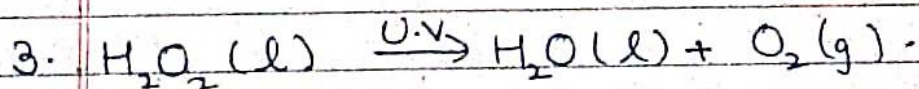
∴ The reaction observed here is Combination reaction.



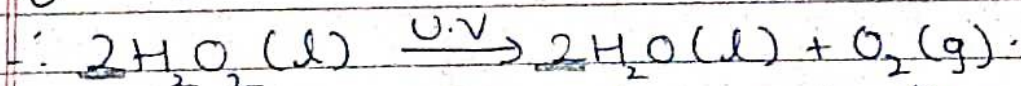
	R	P
Ti	1	1
Cl	4	$2 \times 2 = 4$
Mg	$1 \times 2 = 2$	1

∴ $\text{TiCl}_4\text{(l)} + 2\text{Mg (s)} \rightarrow \text{Ti (s)} + 2\text{MgCl}_2\text{(s)}$ is the eqⁿ.

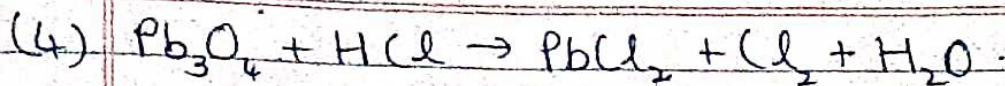
∴ The reaction observed here is :- Double Displacement Reaction.



	R	P
H	$2 \times 2 = 4$	$2 \times 2 = 4$
O	2	3



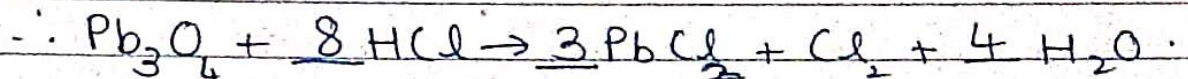
∴ The reaction observed here is Decomposition Reaction.



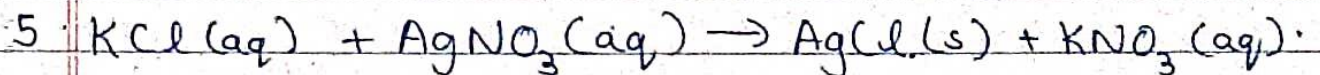
	R	P
Pb	3	1
O	4	1 × 4
H	1 × 8	2 × 4
Cl	1	4



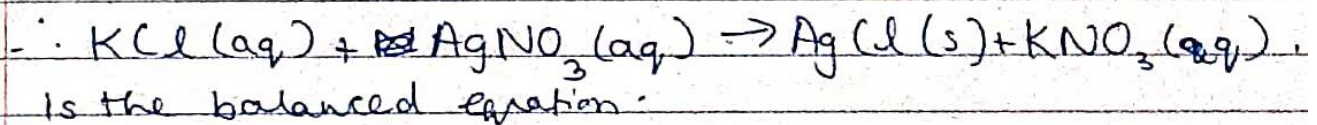
	R	P
Pb	3	3
O	4	4
H	8	8
Cl	8	8



$\therefore$ This is a Redox Reaction.



K	1	1
Cl	1	1
Ag	1	1
NO_3	1	1



$\therefore$ Displacement Reaction has occurred here.